

Building a Cloud-Based Inventory Dashboard with AWS: A Teaching Guide

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Introduction

Efficient inventory tracking is essential for businesses to minimize stockouts and optimize replenishment. This guide demonstrates how to create a cloud-based Supply Chain Dashboard using AWS services, offering an automated way to monitor inventory levels through a simple web interface.

This hands-on project is particularly valuable for business educators aiming to integrate applied technology into their curriculum. By combining operations management, cloud infrastructure, and data visualization, this exercise allows students—regardless of technical background—to explore how modern businesses can gain real-time insight into inventory. It serves as a capstone-style assignment that can support coursework in supply chain management, analytics, or IT strategy.

Tasks Overview

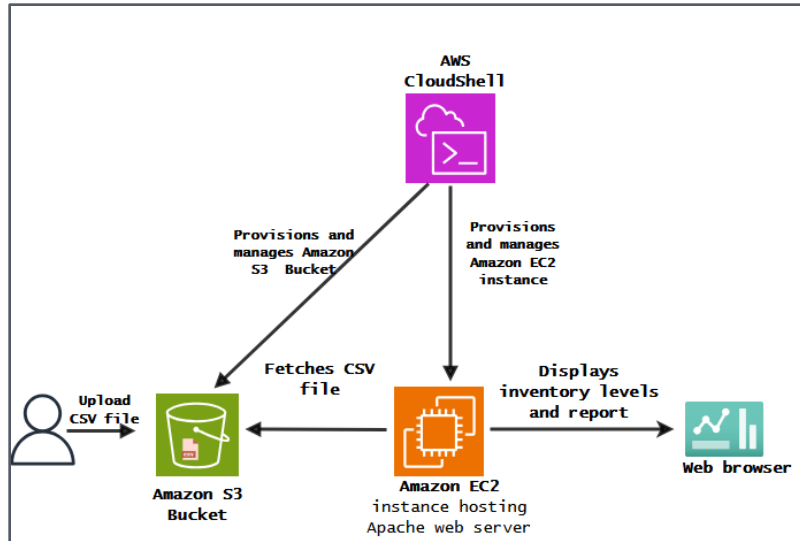
In this tutorial, you will:

1. Create an Amazon S3 bucket to store inventory data.
2. Configure public read access for the S3 bucket (for demonstration purposes only).
3. Upload inventory data to the S3 bucket.
4. Create an EC2 instance and set up an Apache web server.
5. Deploy a PHP-based dashboard to visualize inventory data.
6. Configure security settings for the EC2 instance.
7. Access and test the inventory dashboard.
8. Clean up AWS resources after completion.

Estimated time for completion: 45 - 60 minutes

Architecture Overview

This cloud-based inventory dashboard leverages AWS services to create an efficient solution for storing, processing, and displaying inventory data. The architecture is designed to provide businesses with real-time insights into their inventory levels through a user-friendly web interface.



Key Architecture Features:

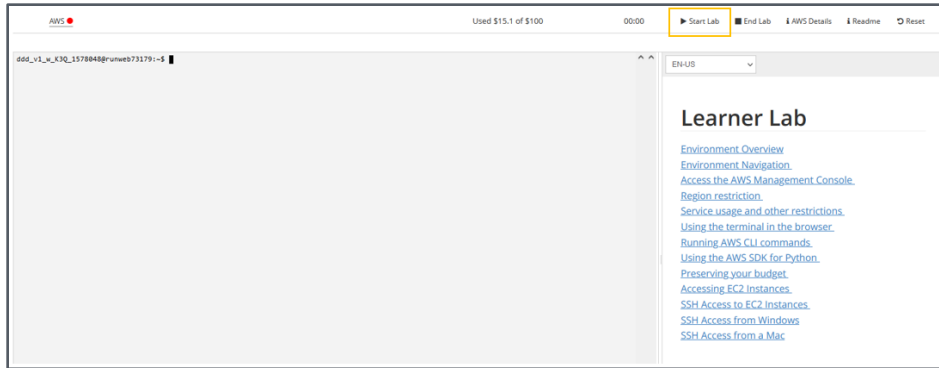
1. AWS CloudShell provisions and manages both the Amazon S3 bucket and the Amazon EC2 instance.
2. The EC2 instance runs an Apache web server that hosts the PHP-based dashboard application.
3. This architecture enables a seamless flow from data upload to visualization, facilitating efficient inventory management.

By implementing this cloud-based inventory dashboard, business educators and students can gain hands-on experience in modern inventory management techniques. This practical exposure enhances skills in operations analytics and inventory optimization within a cloud environment.

Prerequisites

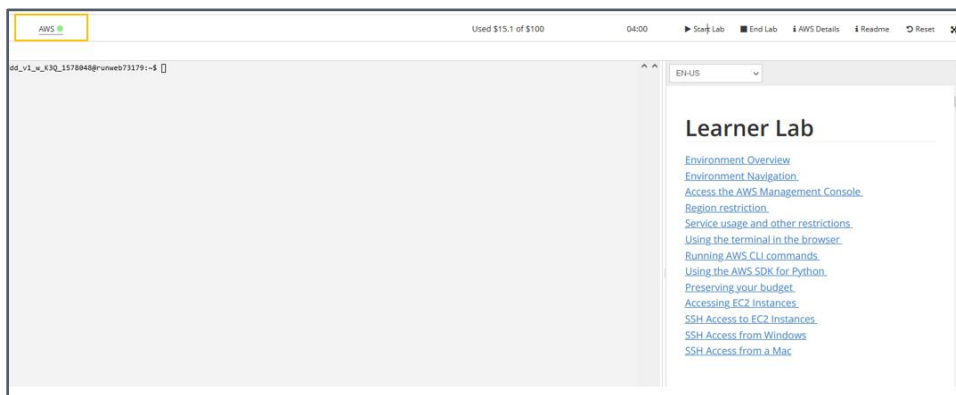
AWS Environment Setup

- **Access to the [AWS Academy Learner Lab \(LL\)](#) course:** From the Student View of the Learner Lab, click the "Start Lab" button in the upper right corner of the interface, as shown in the screenshot below:



Note: For information on how to view a Learner Lab course in student view, please review **How to view a Learner Lab in Student View** section from [AWS Academy Learner Lab Educator Guide](#).

- After starting the lab, wait until the AWS icon turns green in the upper left corner of the interface, as shown below:



Then click on the AWS icon to display the AWS Management console.

Required Files

Download and extract Lab_Files.zip accompanying this article to your local machine. The zip folder contains:

- inventory_data.csv (inventory sample data)
- dashboard.php (web dashboard script)

AWS CloudShell Access and File Setup

Access AWS CloudShell:

- In the AWS Management Console, click the CloudShell icon in the top navigation bar (shell symbol) •
- Or click the "Search" bar and type "CloudShell"
- Wait for the CloudShell environment to initialize

Upload Files to CloudShell:

Using Upload Button

- Click the "Actions" button (⚙) in CloudShell
- Select "Upload file"
- Choose the required files from your local machine:
 - inventory_data.csv
 - dashboard.php

Verify File Upload:

```
# List uploaded files
```

```
ls -l
```

```
# Check file contents if needed
```

```
head inventory_data.csv
```

This lab exercise uses these files to create a web-based inventory dashboard on AWS cloud infrastructure.

Note - While this lab uses simplified security settings for learning purposes, it's important to understand that in real business environments, AWS implementations require careful attention to security best practices and data protection policies. This hands-on exercise provides a foundation for understanding cloud security concepts, which you'll need to expand upon for actual business deployments.

Implementation

Store Inventory CSV Files in S3

This implementation uses public access for demonstration purposes. In a production environment, you should enable S3 bucket versioning, implement server-side encryption, configure access logging, and use VPC endpoints for private access.

Step 1: Create an S3 Bucket

Run in AWS CloudShell

```
# Bucket name must be globally unique. Add random numbers if needed (e.g.,  
supply-chain-inventory-bucket-0256).
```

```
aws s3 mb s3://supply-chain-inventory-bucket
```

Step 2: Disable Block Public Access

Run in AWS CloudShell

```
aws s3api put-public-access-block --bucket supply-chain-inventory-bucket --  
public-access-block-configuration  
"BlockPublicAcls=false,IgnorePublicAcls=false,BlockPublicPolicy=false,Restric  
tPublicBuckets=false"
```

Note: This configuration grants public read-only access to view and download only the specified inventory data file in the S3 bucket. While this setup is suitable for learning purposes, in a production environment you should use more secure methods like IAM roles for EC2, signed URLs, or CloudFront with OAI to restrict access to your data.

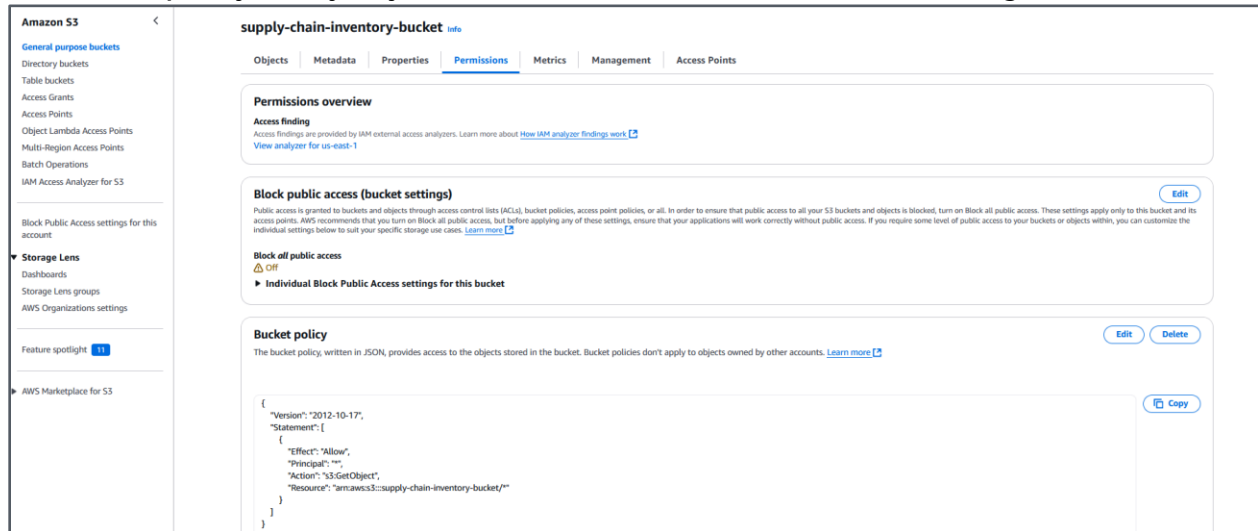
Step 3: Grant Public Read Access

Run in AWS CloudShell

```
aws s3api put-bucket-policy --bucket supply-chain-inventory-bucket --policy '{  
  "Version": "2012-10-17",  
  "Statement": [  
    {  
      "Effect": "Allow",  
      "Principal": "*",  
      "Action": "s3:GetObject",  
      "Resource": "arn:aws:s3:::supply-chain-inventory-bucket/*"  
    }  
  ]  
}'
```

This policy allows anyone to read files from the S3 bucket, ensuring that the PHP script can access `inventory_data.csv` without authentication.

After adding the above policy, the Bucket Policy section of the Permissions tab should reflect the policy that you just added, and should match the following screenshot:



Step 4: Upload Sample Inventory Data

Run in AWS CloudShell

```
aws s3 cp inventory_data.csv s3://supply-chain-inventory-bucket/
```

Note: Before running this command, ensure that `inventory_data.csv` is in your current CloudShell directory. You can upload the file to CloudShell using the file upload feature.

Create a Key Pair for EC2

To securely connect to your EC2 instance (i.e., your cloud-based server), you'll need a key pair. Run the following commands in AWS CloudShell:

```
aws ec2 create-key-pair --key-name SupplyChainKey --query 'KeyMaterial' --
output text > SupplyChainKey.pem
chmod 400 SupplyChainKey.pem
```

Launch an Apache Web Server on EC2

Step 1: Launch EC2 Instance

Run in AWS CloudShell

Before running this command, replace `ami-xxxxxxxxxxxxxxxxxx` with a valid AMI ID from your AWS region:

```
aws ec2 run-instances --image-id ami-xxxxxxxxxxxxxxxxxx --count 1 --instance-type t2.large \
--key-name SupplyChainKey --security-groups default --tag-specifications \
'[{"ResourceType":"instance","Tags":[{"Key":"Name","Value":"SupplyChainServer"}]]'
```

Note: To find a valid AMI ID: Open the EC2 console > Images > AMI Catalogue > Quick Start AMIs tab > Select Amazon Linux 2 AMI (64-bit x86) > Copy AMI ID. Tip: Choose AMIs labeled "Free tier eligible" to minimize costs.

Step 2: Connect to your EC2 using SSH

Run in AWS CloudShell

```
ssh -i SupplyChainKey.pem ec2-user@<INSTANCE_PUBLIC_IP>
```

Replace `<INSTANCE_PUBLIC_IP>` with your EC2 instance's public IP address.

Before connecting, ensure:

1. The EC2 instance's security group allows inbound SSH (port 22) from your IP address.
2. The security group also allows inbound HTTP (port 80) from any IP address (0.0.0.0/0).

These settings are crucial for:

- Connecting to your EC2 instance via SSH
- Allowing public access to the dashboard from the internet

Once connected, update the system and install Apache and PHP.

Step 3: Install Apache & PHP

Run inside EC2:

```
sudo yum update -y
sudo yum install -y httpd php php-mysql
sudo systemctl start httpd
sudo systemctl enable httpd
```

Build a PHP Dashboard for Stock Tracking

1. Upload `dashboard.php` to CloudShell.

2. Transfer `dashboard.php` from CloudShell to the EC2 instance:

```
scp -i SupplyChainKey.pem dashboard.php ec2-user@[INSTANCE_PUBLIC_IP]:/home/ec2-user/
```

3. Connect to the EC2 instance:

```
ssh -i SupplyChainKey.pem ec2-user@[INSTANCE_PUBLIC_IP]
```

Replace `[INSTANCE_PUBLIC_IP]` with your EC2 instance's public IP address in the above commands.

4. Move the file to Apache's directory and set appropriate permissions:

```
sudo mv /home/ec2-user/dashboard.php /var/www/html/  
sudo chown apache:apache /var/www/html/dashboard.php  
sudo chmod 644 /var/www/html/dashboard.php
```

Understanding the `dashboard.php` Code Structure

Take some minutes to review the `dashboard.php` code. This code serves as the "brain" of the dashboard—it reads inventory data and turns it into charts and a report.

Here's a simplified breakdown:

- **Fetching Data:** Reads the CSV file from S3 using a public URL
- **Processing Data:** Extracts headers, calculates inventory metrics
- **Generating Charts:** Uses Chart.js for visualizations (bar and line charts)
- **Displaying Results:** Shows inventory in an interactive table and graphs

Note: Even if you're not familiar with PHP, you can understand how this code transforms your data into visual information.

Access the Dashboard

To view your dashboard, open a web browser and enter your EC2 instance's public IP address followed by `/dashboard.php`.

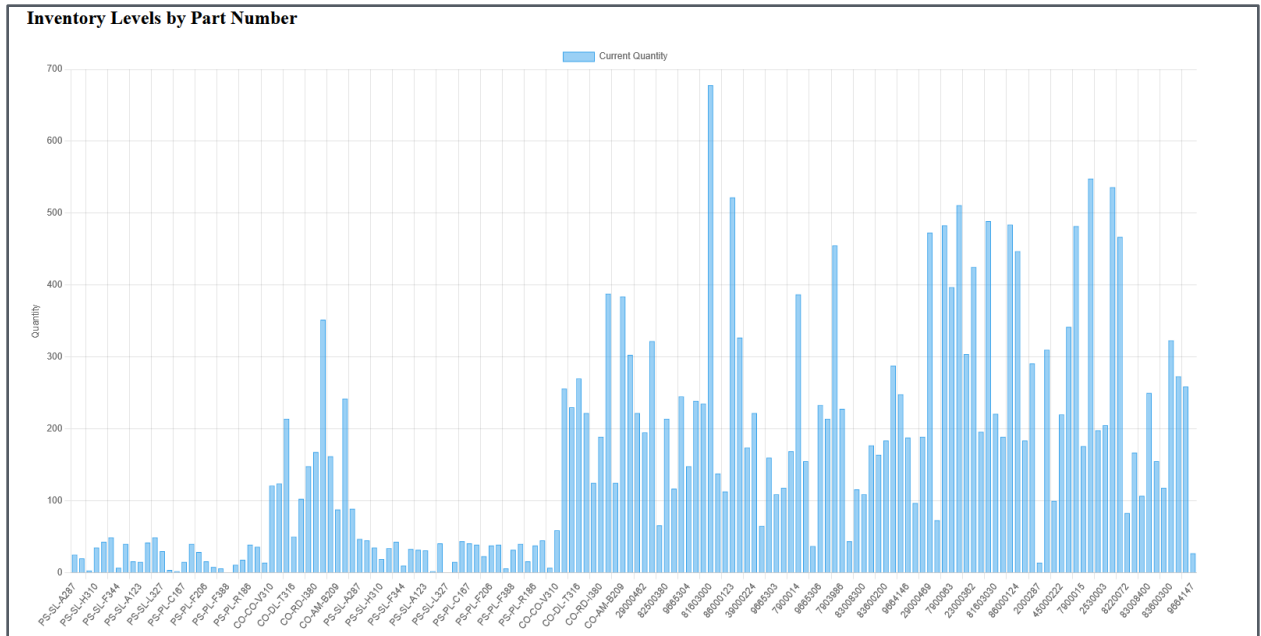
```
http://<INSTANCE_PUBLIC_IP>/dashboard.php
```

Your Dashboard Features

The dashboard displays three main components to help you monitor your inventory:

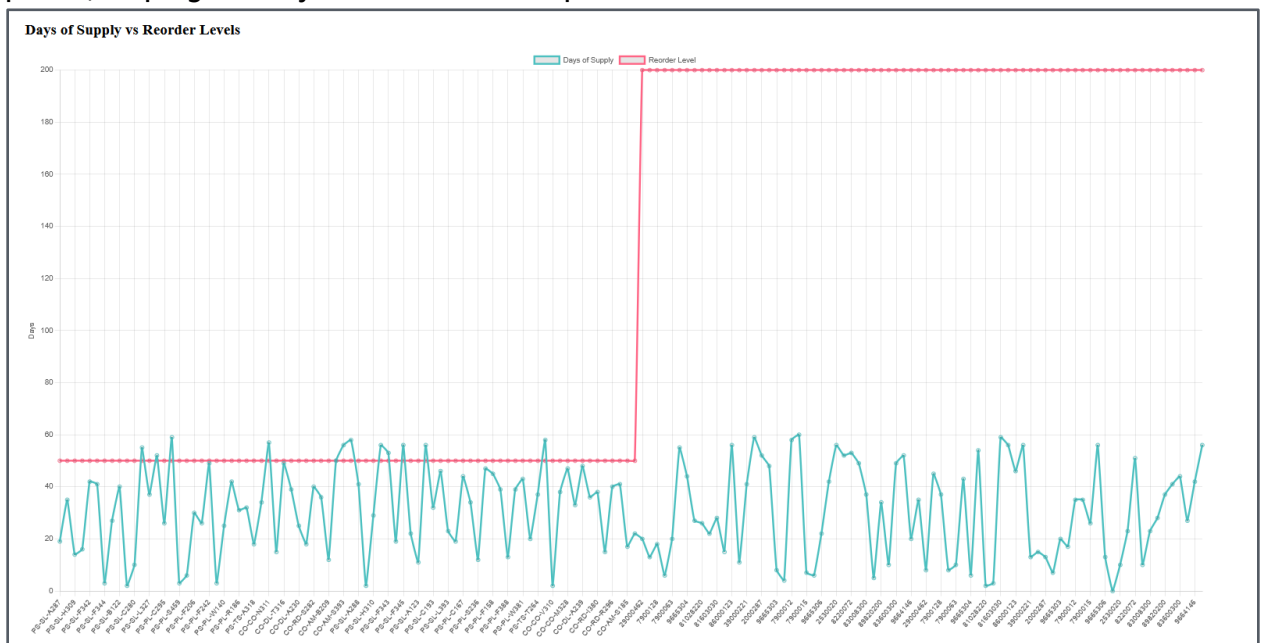
1. Inventory Levels by Part Number

This bar chart shows the current quantity of each part in your inventory, making it easy to visualize stock levels across different items.



2. Days of Supply vs Reorder Levels

This chart compares your current inventory duration against predetermined reorder points, helping identify items that need replenishment.



3. Detailed Inventory Report

A comprehensive table view provides detailed information about each part, including current quantities, reorder levels, and other key metrics.

Detailed Inventory Report						
Part Number	Location	Quantity	Days of Supply	Reorder Level	Value	Lead Time
PS-SL-A287	LT-1	25 EA	19	50	31250 USD	30
PS-SL-A288	LT-1	20 EA	35	50	25000 USD	30
PS-SL-H009	LT-1	3 EA	14	50	3750 USD	30
PS-SL-H010	LT-1	35 EA	16	50	43750 USD	30
PS-SL-F342	LT-1	43 EA	42	50	53750 USD	30
PS-SL-F343	LT-1	49 EA	41	50	61250 USD	30
PS-SL-F344	LT-1	7 EA	3	50	8750 USD	30
PS-SL-F345	LT-1	40 EA	27	50	50000 USD	30

Testing the Setup

- Ensure Apache is running
- Verify PHP script loads correctly
- Confirm charts and tables display inventory

Clean Up Resources

To avoid charges, delete the resources when finished.

Run in AWS CloudShell:

```
aws ec2 terminate-instances --instance-ids <INSTANCE_ID>
aws s3 rb s3://supply-chain-inventory-bucket -force
```

After completing this lab exercise, verify all sensitive data is removed from the S3 bucket. Review CloudTrail logs for any unauthorized access, document any security incidents during the exercise, and ensure all public access permissions are removed.

After completing this tutorial, students should be able to:

- Explain how AWS supports supply chain operations
- Use Amazon S3 and EC2 effectively
- Interpret data visualizations in operational contexts
- Understand and troubleshoot basic cloud application issues